JOSEPH BRUNET

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EDUCATION

École des Mines de Saint-Étienne, France *Nov 2017 - Mar 2021*
Ph.D. in Biomechanics

Paris Descartes University, France *Sep 2016 - Oct 2017*
Master of Science - MS, Biomedical Engineering - Major in Biomechanics

École Nationale Supérieure d'Arts et Métiers, France *Sep 2014 - Oct 2017*
Master's Degree in Mechanical Engineering

RESEARCH EXPERIENCE

Interim Director, HOAHub | University College London (UCL), *May 2025 - Present*
Funded by the Chan Zuckerberg Initiative

Appointed interim director of the Human Organ Atlas Hub (HOAHub), an interdisciplinary UCL-ESRF research infrastructure delivering open-access, high-resolution 3D imaging datasets of intact human organs using Hierarchical Phase-Contrast Tomography (HiP-CT).

- Provide scientific and strategic leadership for HOAHub and its international imaging programme (HiP-CT)
- Coordinate operations across ESRF and eight co-investigator organisations to deliver milestones and timelines
- Manage collaborations with 30+ international research groups using HOAHub infrastructure
- Oversee public data dissemination via the Human Organ Atlas platform (human-organ-atlas.esrf.eu) for research, education, and clinical translation
- Represent HOAHub at international events, oversee reporting and funding coordination, and lead working groups on imaging, biomechanics, and integration with clinical modalities

Research Associate | University College London (UCL), *Sep 2023 - Present*
Funded by the Chan Zuckerberg Initiative

Research Associate in ultra-high resolution quasi-dynamic X-ray imaging of human organs and joints using synchrotron tomography

- Visiting scientist at the European Synchrotron Radiation Facility (ESRF)
- Locally lead and co-ordinate the development of the HiP-CT technique at ESRF, performing HiP-CT scans using the latest developments
- Managing other UCL staff and students at ESRF (1 PDRA and, act as local UCL manager for ESRF staff funded by UCL (currently 4 staff))
- Ownership of the local activities of the HiP-CT project at ESRF, coordinating interactions with ESRF Staff, the 8 other Co-Investigator organisations, and the over 30 supporting groups
- Technical leadership. To develop and apply imaging analysis techniques, especially digital volume correlation, to interpret quasi-dynamic biological system behaviour
- Coordinate the local budget of UCL at ESRF for the HiP-CT related projects
- Co-supervision PhD/MSc students, acting as local supervisor whilst they are doing placements at ESRF

Postdoctoral researcher | University College London (UCL), *Sep 2021 - Sep 2023*
Funded by the Chan Zuckerberg Initiative

Research Fellow in ultra-high resolution quasi-dynamic X-ray imaging of human organs and joints using synchrotron tomography

- Visiting scientist at the European Synchrotron Radiation Facility (ESRF)
- Preparation of biological human and animal samples for quasi-dynamic biological scanning
- Perform Hierarchical Phase-Contrast Tomography (HiP-CT) scans at the European Synchrotron Radiation Facility (ESRF)
- Development of a quasi-dynamic experimental setup to perform in situ testing on coronary stent expansion in a complete heart
- Development and application of imaging analysis techniques, in particular digital volume correlation, to interpret quasi-dynamic biological system behaviour

PhD Researcher | École des Mines de Saint-Étienne, France *Nov 2017 - Sept 2021*
Funded by the European Research Council
Advisors: Pierre Badel, Éric Maire

Thesis: Understanding the mechanisms of aortic dissection: finite element modeling and *in situ* experimentation with X-ray tomography

- Collection and manipulation of arterial tissue from pigs and rabbits
- Tensile and peeling tests to assess the elastic and fracture properties of the tissue
- Development of a tension-inflation device fitting inside an X-ray tomography setup and its computer interface
- In situ 3D imaging of the dissection of pressurized arterial segments by X-ray microtomography and synchrotron-based phase-contrast imaging
- Modeling of the observed failure phenomena using the extended finite element method

Research intern | Medtronic, France *Jul 2017 - Nov 2017*
Experimental and numerical study on meshes for abdominal hernia repair

- Uniaxial, biaxial, and indentation tests on different type of textiles
- Modeling the observed mesh behaviors with a nonlinear, anisotropic, plastic model

M.S. Researcher | École des Mines de Saint-Étienne, France *Jan 2017 - Jul 2017*
Advisor: Pierre Badel

Thesis: Investigation of shear delamination during rupture of arterial medial tissue using cohesive numerical model

PUBLICATIONS

1. Docter, D., **J. Brunet**, V. Sabarigirivasan, G. Lemmens, B. Tielemans, J. Sporring, H. Dejea, T. Urban, J. Purzycka, R. Gorter, J. Huirne, V. Michels, J. Hanemaaijer-van der Veer, A. Bellier, D. Stansby, M. Ackermann, R. D. Buelow, D. D. Jonigk, J. Jacob, J. Hagoort, S. Verleden, A. Cook, C. Walsh, P. Tafforeau, P. Lee & B. de Bakker (2026). Hierarchical Phase-Contrast Tomography Imaging: Applicability in biomedical research, *bioRxiv*. Preprint. doi: 10.64898/2026.01.15.699614.

2. Dejea, H., **J. Brunet**, T. Urban, C. Berruyer, A. Mirone, C. Jarnias, A. Bellier, C. L. Walsh, P. D. Lee & P. Tafforeau (2026). Evolution of Hierarchical Phase-Contrast Tomography on the European Synchrotron beamlines BM05 and BM18: a whole adult human brain imaging case study, *bioRxiv*. Preprint. doi: 10.64898/2026.01.15.699653.

3. Veerle M. W. Michels, Adam Szmul, Joseph Jacob, Hector Dejea, Bernadette S. de Bakker, Judith A. F. Huirne & **HOAHub Uterine Consortium**. (2026) Mapping the arterial vascular network in an intact human kidney using hierarchical phase-contrast tomography, *Angiogenesis*. 29(3). doi: 10.1007/s10456-025-10004-w.

4. Urban, T., **J. Brunet**, M. Didziokas, M. Chourrout, A. Keenlyside, K. Orsine Murta Dias, A. Mirone, A. Bellier, H. Dejea, L. Lammalle, B. de Bakker, H. Petrushkin, M. Moazen, C. L. Walsh, P. D. Lee & P. Tafforeau (2025). Resolving near-micron scale features within a whole sheep head using Hierarchical Phase-Contrast Tomography, *bioRxiv*. Preprint. doi: 10.64898/2025.12.15.694379.

5. Sabarigirivasan, V., **J. Brunet**, H. Dejea, A. Crucean, A. Jegatheeswaran, C. Capelli, E. Pajaziti, T. Urban, J. Purzycka, P. Tafforeau, C. L. Walsh, P. D. Lee & A. C. Cook (2025). The heterogeneous nature of atrioventricular conduction tissues in Tetralogy of Fallot demonstrated by hierarchical phase-contrast tomography, redefining the anatomic substrate, *bioRxiv*. Preprint. doi: 10.64898/2025.12.04.692356.

6. Sharma, A., Evans, L. A. E., Bourne, L. E., Chen, J., Parmenter, A. L., **J. Brunet**, Madi, K., Marussi, S., Tafforeau, P., Pitsillides, A. A., Lee, P. D. & Staines, K. A. (2025). In situ profiling of nanoscale displacements uncovers mechano-architectural predictors of osteoarthritis emergence, *bioRxiv*. Preprint. doi: 10.64898/2025.12.04.692296.

7. **J. Brunet**, C. L. Walsh, P. Tafforeau, A. C. Cook, K. Engel, A. Bellier, C. Werlein, F. Lnger, J. Jacob, M. Ackermann, D. D. Jonigk, P. D. Lee (2025). Hierarchical Phase-Contrast Tomography: Advancing High-Resolution Imaging of Intact Human Organs, *Grays Anatomy, 43rd edition*. Book chapter.

8. Newham, E., Parmenter, A. L., Chen, J., Disney, C., Snow, T., **J. Brunet**, Vinci, V., Drnec, J., Sherratt, M. J., Hoyland, J. A., Bellier, A., Bay, B. K., Terrill, N., Lee, P. D. & Gupta, H. S. (2025) Multimodal X-ray imaging reveals hierarchical fibre mechanics, *bioRxiv*. Preprint. doi: 10.1101/2025.09.19.677294.

9. **J. Brunet**, Chestnutt, L., Chourrout, M., Dejea, H., Sabarigirivasan, V., Lee, P. D. & Cook, A. C. (2025) Cardiotensor: A Python Library for Orientation Analysis and Tractography in 3D Cardiac Imaging, *arXiv*. Preprint. doi: 10.48550/arXiv.2508.07476.

10. Walsh, C. L., **Brunet, J.**, Stansby, D., Gaisn, G., Zhou, Y., Ackermann, M., ... Lee, P. D. (2025). The Human Organ Atlas. bioRxiv, 2025-07.

11. Rahmani, S., Jafree, D. J., Lee, P. D., Tafforeau, P., **J. Brunet**, Nandanwar, S., Jacob, J., Bellier, A., Ackermann, M., Jonigk, D. D., Shipley, R. J., Long, D. A. & Walsh, C. L. (2025) Mapping the arterial vascular network in an intact human kidney using hierarchical phase-contrast tomography, *npj Imaging*. 3(1), p.39. doi: 10.1038/s44303-025-00090-2.

12. Zhou, Y., Aslani, S., Javanmardi, Y., **J. Brunet**, Stansby, D., Carroll, S., Bellier, A., Ackermann, M., Tafforeau, P., Lee, P. D. & Walsh, C. L. (2025) Multiscale segmentation using hierarchical phase-contrast tomography and deep learning, *bioRxiv*. Preprint. doi: 10.1101/2025.05.15.654263.

13. Mirone, A., **J. Brunet**, Boistel, R., Sowinski, M., Berruyer, C., Payno, H., Boller, E., Paleo, P., Walsh, C. L., Lee, P. D. & Tafforeau, P. (2025) Eikonal phase retrieval: Unleashing the fourth generation sources' potential for enhanced propagation-based tomography on biological samples, *Journal of Synchrotron Radiation*, 32(5). doi: 10.1107/S1600577525005223.

14. Yagis, E., Aslani, S., Jain, Y., Zhou, Y., Rahmani, S., **J. Brunet**, Bellier, A., Werlein, C., Ackermann, M., Jonigk, D., Tafforeau, P., Lee, P. D. & Walsh, C. (2024) Deep learning for 3D vascular segmentation in hierarchical phase contrast tomography: a case study on kidney, *Scientific Reports*, 14(1), p. 27258. doi: 10.1038/s41598-024-77582-5.

15. **J. Brunet**, Walsh, C., Tafforeau, P., Dejea, H., Cook, A., Bellier, A., Engel, K., Jonigk, D. D., Ackermann, M. & Lee, P. D. (2024) Hierarchical phase-contrast tomography: A non-destructive multiscale imaging approach for whole human organs, *Developments in X-Ray Tomography XV*, Vol. 13152, pp. 1315216. SPIE. doi:10.1117/12.3028717.

16. Jain, Y., Walsh, C. L., Yagis, E., Aslani, S., Nandanwar, S., Zhou, Y., Ha, J., Gustilo, K. S., **J. Brunet**, Rahmani, S., Tafforeau, P., Bellier, A., Weber, G. M., Lee, P. D. & Brner, K. (2024) Vasculature segmentation in 3D hierarchical phase-contrast tomography images of human kidneys, *bioRxiv*. Preprint. doi: 10.1101/2024.08.25.609595.

17. Verleden, S. E., Urban, T., **J. Brunet**, Wen, W., Peeters, D. J. E., Lapperre, T. S., Walsh, C., Tafforeau, P., Dejea, H., Jonigk, D., Ackermann, M., Van Hoorenbeeck, K., Lee, P. D., Jacob, J. & Hendriks, J. M. H. (2024) Imaging three-dimensional airway morphology in congenital pulmonary airway malformation using hierarchical phase-contrast tomography, *American Journal of Respiratory and Critical Care Medicine*. doi: 10.1164/rccm.202403-0509IM.

18. **J. Brunet**, Cook, A. C., Walsh, C. L., Cranley, J., Tafforeau, P., Engel, K., Berruyer, C., Burke OLeary, E., Bellier, A., Torii, R., Werlein, C., Jonigk, D. D., Ackermann, M., Dollman, K. & Lee, P. D. (2024) Multidimensional analysis of the adult human heart in health and disease using hierarchical phase-contrast tomography (HiP-CT), *Radiology*, 312(1). doi: 10.1148/radiol.232731.

19. Xian, R. P., **J. Brunet**, Huang, Y., Wagner, L. W., Lee, D. L., Tafforeau, P. & Walsh, L. C. (2024) A closer look at high-energy X-ray-induced bubble formation during soft tissue imaging, *Synchrotron Radiation*, 31(3), pp. 566577. doi:10.1107/s160057752400290x.

20. **J. Brunet**, Walsh, C. L., Wagner, W. L., Bellier, A., Werlein, C., Marussi, S., Jonigk, D. D., Verleden, E. S., Ackermann, M., Lee, D. P. & Tafforeau, P. (2023) Preparation of large biological samples for high-resolution, hierarchical, multi-modal imaging, *Nature Protocols*, 18, pp. 14411461. doi: 10.1038/s41596-023-00804-z.

21. Ackermann, M., Tafforeau, P., **J. Brunet**, Kamp, J. C., Werlein, C., Khnel, M. P., Jacob, J. et al. (2023) Comment on: Intrapulmonary shunt and alveolar dead space in a cohort of patients with acute COVID-19 pneumonitis and early recovery, *European Respiratory Journal*, 61(2). doi: 10.1183/13993003.02121-2022.

22. **J. Brunet**, Pierrat, B., Adrien, J., Maire, E., Curt, N., Bravin, A., Laroche, N. & Badel, P. (2022) In situ visualization of aortic dissection propagation in notched rabbit aorta using synchrotron X-ray tomography, *Acta Biomaterialia*, 155, pp. 449460. doi: 10.1016/j.actbio.2022.10.060.

23. **J. Brunet**, Pierrat, B. & Badel, P. (2021) A parametric study on factors influencing the onset and propagation of aortic dissection using the extended finite element method, *IEEE Transactions on Biomedical Engineering*, 68(10), pp. 29182929. doi: 10.1109/TBME.2021.3056022.

24. **J. Brunet**, Pierrat, B., Adrien, J., Maire, E., Curt, N. & Badel, P. (2020) A novel method for in vitro 3D imaging of dissecting pressurized arterial segments using X-ray microtomography, *Experimental Mechanics*, 61, pp. 147157. doi: 10.1007/s11340-020-00645-x.

25. **J. Brunet**, Pierrat, B. & Badel, P. (2021) Review of current advances in the mechanical description and quantification of aortic dissection mechanisms, *IEEE Reviews in Biomedical Engineering*, 14, pp. 240255. doi: 10.1109/RBME.2019.2950140.

26. **J. Brunet**, Pierrat, B., Adrien, J., Maire, E. & Badel, P. (2019) A combined experimental-numerical lamellar-scale approach of tensile rupture in arterial medial tissue using X-ray tomography, *Journal of the Mechanical Behavior of Biomedical Materials*, 95, pp. 116123. doi: 10.1016/j.jmbbm.2019.03.028.

INVITED TALKS

1. "Synchrotron imaging and organology, from Paganinis violin to the mapping of human organs" Inaugural lecture delivered at the **25th Annual Meeting of the Francophone Society of Transplantation**, Grenoble, France. Date: December 2nd, 2025

2. "Pushing the limits of HiP-CT: challenges and opportunities in large-volume biological X-ray imaging" Oral presentation delivered at the **Hard X-ray imaging of biological soft tissues symposium 2025**, Crick Institute, Londres, UK. Date: October 13th, 2025

3. "Near-cellular to Whole Organ Quantitative Mapping of the Human Heart Using Hierarchical Phase-Contrast Tomography" Oral presentation delivered at the **BioMedEng25**, Glasgow University, Glasgow, UK. Date: September 5th, 2025

4. "In-Depth Imaging of the Whole Human Adult Heart via Hierarchical Phase-Contrast Tomography" Invitation for oral presentation at **The power of scattering and imaging in understanding and treating disease** conference organized by the LINXS institute, given at Lund University, Lund, Sweden. Date: November 25th, 2024

5. "Overcoming Data Analysis Challenges: Advancing Deep Learning-Based Segmentation in HOAHub Datasets" Invitation for oral presentation at the **Advancing Medical Imaging - Integrating AI and Data Sharing for the Human Organ Atlas Hub** workshop organized by the Center for Quantification of Imaging Data from MAX IV, given at the Technical University of Denmark, Copenhagen, Danemark. Date: November 26th, 2024

6. "Exceptionally detailed Analysis of Whole Human Adult Heart Using Hierarchical Phase- Contrast Tomography" Oral presentation delivered at the **BioMedEng24**, Queen Mary University of London, London, UK. Date: September 7th, 2024

7. "Creating The Human Organ Atlas with HiP-CT" Invitation for oral presentation at the **SPIE Optics + Photonics**, San Diego, United-States. Date: August 23th, 2024

8. "Creating The Human Organ Atlas with HiP-CT" Invitation for oral presentation at the **Big Data and Big Science** workshop organized by University of the Witwatersrand, Johannesburg, South-Africa. Date: May 8th, 2024

9. "Human Organ Atlas project : High-resolution imaging of whole human organs and application on COVID-19" Invitation for oral presentation at the **XXVImes Journes Francophones de Virologie**, Brussels, Belgium. Date: April 11th, 2024

10. "In situ quantification of the mechanical behavior of arteries using synchrotron X-ray tomography" Invitation for oral presentation at the **GDR Matriaux Fibreux - Caractrisation, modlisation et optimisation (GDR CNRS FIBMAT)**, Lyon, France. Date: December 1st, 2023

11. "Hierarchical Phase-Contrast Tomography for High-Resolution imaging of Ex Vivo Whole Adult Human Heart in Health and Disease" Oral presentation delivered at the **BioMedEng23**, Swansea University, Swansea, UK. Date: September 14th, 2023

12. "BM18, the new ESRF-EBS beamline for multiresolution phase-contrast tomography on large samples : past, present and future" Invitation for key-lecture presentation at the "High-resolution imaging in correlative workflows" workshop organized by Zeiss given at the **Technical University of Munich**, Munich, Germany. Date: July 3rd, 2023

13. "X-ray imaging at BM18 - Human Organ atlas project" Seminar given at **Ecole des Mines de Saint-Etienne**, Saint-Etienne, France. Date: July 3rd, 2023

14. "Imaging of whole human organs using Hierarchical Phase-Contrast Tomography (HiP-CT)" Invitation for oral presentation at the **6th Annual Workshop on Advances in X-ray imaging**, Harwell, UK. Date: June 22nd, 2023

15. "Human Organ Atlas project: Whole Human Organs Imaged in Synchrotrons" Invitation for oral presentation at workshop (**"3D image analysis From Foundation to Vision"**) at the Center for Quantification of Imaging Data from MAX IV (QIM), Copenhagen, Denmark. Date: March 10th, 2023

16. "Imaging of whole human organs using Hierarchical Phase-Contrast Tomography (HiP-CT)" Oral presentation delivered at the ImagingBioPro Network Workshop (Leveraging Next-Gen Multiscale Imaging at Central Facilities for Biomedical Engineering Research) at the **BioMedEng22**, University College London, London. Date: September 9th, 2022

17. "In situ quantification of the mechanical behavior of coronary arteries during angioplasty using synchrotron X-ray tomography in intact porcine hearts" Oral presentation delivered at the **BioMedEng22**, University College London, London. Date: September 9th, 2022

18. "Imaging of whole human organs using Hierarchical Phase-Contrast Tomography (HiP-CT)" Seminar given at **Witwatersrand University**, Johannesburg, South Africa. Date: September 6th, 2022

19. "High-resolution imaging of the whole human heart using synchrotron x-ray tomography" Oral presentation delivered at the **5th International Conference on Tomography of Materials and Structures (ICTMS)**, Grenoble, France. Date: July 25th, 2022

20. "3D characterization of crack propagation during the onset of a dissection using X-ray microtomography on pressurized aortic segments" Oral presentation delivered at the **26th Congress of the European Society of Biomechanics**, Online. Date: July 13th, 2021

21. "Hierarchical imaging of complete human organs, how the EBS changed the game" Oral presentation at the **Science Day Symposium**, ESRF, France. Date: March 21st, 2022

22. "Investigation of notch propagation in an in-vitro dissection model using X-ray microtomography" Oral presentation delivered at the **45th Congress of the Societe de Biomecanique**, Online. Date: October 27th, 2020

23. "A numerical design of the experiment approach to understand aortic dissection onset and propagation" Oral presentation delivered at the **44th Congress of the Societe de Biomecanique**, Poitiers, France. Date: October 29th, 2019

24. "Chairman of the session 'Macro-scale biofluids'" **44th Congress of the Societe de Biomecanique**, University of Poitiers, France. Date: October 28th, 2019

25. "A new approach combining experiment and numerical simulation using a cohesive interface to model tensile failure in arterial medial tissue at the meso-scale" Poster presentation delivered at the **8th World Congress of Biomechanics**, Dublin, Ireland. Date: July 9th, 2018

26. "Characterization and modeling of rupture in arterial medial tissue under tension from in situ experiments with X-ray tomography" Oral presentation delivered at the **15th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering**, Lisbon, Portugal. Date: March 28th, 2018

GRANTS

- 2025 Biomedical Challenge Beamtime (HOAHub) **ca. £40,000 worth of beamtime, co-PI**
- 2024 - Feasibility beamtime (HOAHub) - **ca. £20,000 worth of beamtime, PI**
- 2024 Human Organ Atlas Hub (HOAHub) **ca. £1,000,000 worth of beamtime, Co-investigator**

- 2023 HiP-CT beamtime MD-1290 (12 beamtimes) **ca. £1,000,000 worth of beamtime, Co-investigator**
- 2022 Proof of Concept Awards (PoCAs) delivered by ImagingBioPro Network. **£25,000 PI**
- 2020 - ESRF Beamtime LS-2904 - **ca. £80,000 PI**

TRAVEL GRANTS

- 2025 CZI Imaging the Future 2025: Focusing on Biology in Action, San Jose, US **£2,000**
- 2024 CIFAR MacMillan Multiscale Human Spring 2025, Geneva, Switzerland **£1,500**
- 2024 CZI Imaging the Future 2024: Life Illuminated, Austin, US **£2,000**
- Travel Grants for Invited Talks (6 - equivalent to approx. **£3600**) *2018 - 2024*
  Awarded travel grants to present research at international conferences or workshops

TEACHING EXPERIENCE

Co-Supervisor of 2 PhD student, UCL *2024 - 2025*
Projects: Coronary arteries mechanical characterization and modelling

Correction of 3rd-year project thesis - UCL *2023 - 2025*
Correction for the Cardiovascular Science MSc

Co-Supervisor of 6 MS students, ESRF/UCL *2021 - 2023*
Projects: Mechanical testing / Image processing and analysis / Deep learning

Provide training to PhD students and Post docs at ESRF *2021 - 2023*
Subjects: sample preparation, scan acquisition, image reconstruction

Supervisor of 2 MS students, École des Mines de Saint-Étienne *2018 - 2020*
Industrial and research projects

Teaching Assistant, École des Mines de Saint-Étienne (140 hours) *2018 - 2020*
Courses: Experimental mechanics, Finite element method

Tutor for students in difficulty, Saint-Exupery middle school, Macon *2015 - 2016*
Courses: Mathematics and Physics

AWARDS

1st prize poster competition (£300). CiET Mid-term review, University College London. March, 2024

Camille Mathieu Award (5000€). Handitecam, Arts et Mtiers. December, 2017

PROFESSIONAL AND INSTITUTIONAL ROLES

UCL representative at ESRF *2023 - 2025*
Locally lead and coordinate research activities, provide assistance to new staff for settling in, facilitate collaboration between UCL and ESRF teams, and ensured smooth operational workflows.

Responsible for health and safety compliance of the project at ESRF *2021 - 2025*
Handling human organ samples, dealing with substances like ethanol, formalin, and the inherent risks of X-ray radiation

Guest Lecturer in Engineering Schools *2022 - 2024*
Delivered presentations on current research and academic opportunities in the sciences to undergraduate engineering students. Aims to inspire students to consider careers in academia.

PhD student representative *2018 - 2020*
Board of the Doctoral School of Science, Engineering and Health of the University of Lyon

PhD student representative *2018 - 2020*
Board of SAINBIOSE laboratory (INSERM)

President of the Saint-Étienne Doctoral Students' Club *2018 - 2019*

PROFESSIONAL AFFILIATIONS

Member of European Society of Biomechanics *2020 - Present*
Member of French Society of Biomechanics *2020 - Present*

REFERENCES

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